

```
1 using CustomProjectRPG.Entities;
2 using System;
3 using System.Collections.Generic;
4 using System.Linq;
5 using System.Text;
6 using System.Threading.Tasks;
7 using System.Text.Json;
8 using System.Text.Json.Serialization;
9 using CustomProjectRPG.Interfaces;
10
11 namespace CustomProjectRPG
12 {
13     public class InventoryComponent: ISerializeJSON<InventoryComponent>
14     {
15         private List<Item> _items;
16         private int _maxSlots;
17
18         public InventoryComponent(int maxSlots = 5)
19         {
20             _items = new List<Item>();
21             _maxSlots = maxSlots;
22         }
23
24         public IReadOnlyList<Item> GetContents()
25         {
26             return _items.AsReadOnly();
27         }
28
29         public bool Add(Item item)
30         {
31             if (item == null)
32             {
33                 throw new ArgumentNullException(nameof(item));
34             }
35
36             if (_items.Count >= _maxSlots)
37             {
38                 return false;
39             }
40
41             _items.Add(item);
42             return true;
43         }
44
45         public bool Remove(Item item)
46         {
47             if (item == null)
48             {
49                 throw new ArgumentNullException(nameof(item));
```

```
50     }
51
52     return _items.Remove(item);
53 }
54
55 public bool HasItem(string itemId)
56 {
57     if (string.IsNullOrEmpty(itemId))
58     {
59         throw new ArgumentException("Item ID cannot be null or empty.");
60     }
61
62     foreach (Item item in _items)
63     {
64         if (item.Id == itemId)
65         {
66             return true;
67         }
68     }
69
70     return false;
71 }
72
73
74 public string Serialize()
75 {
76     return JsonSerializer.Serialize(_items);
77 }
78
79 public void Deserialize(string json)
80 {
81     if (string.IsNullOrEmpty(json)) return;
82     var items = JsonSerializer.Deserialize<List<Item>>(json);
83     _items = items ?? new List<Item>();
84 }
85
86 }
87
88 }
89
```